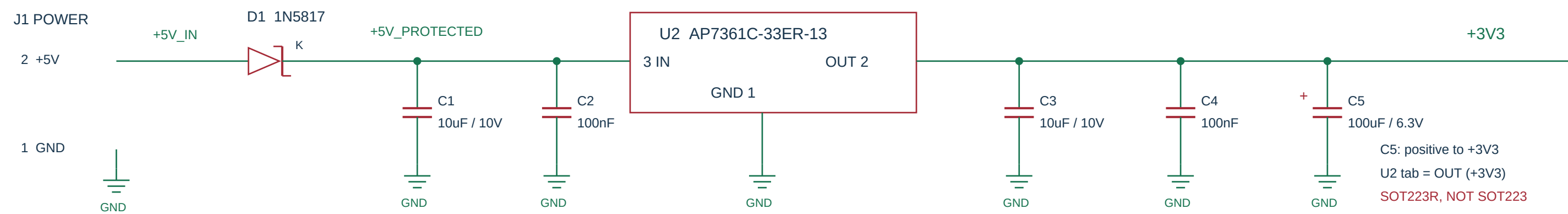


4VRS DISPLAY

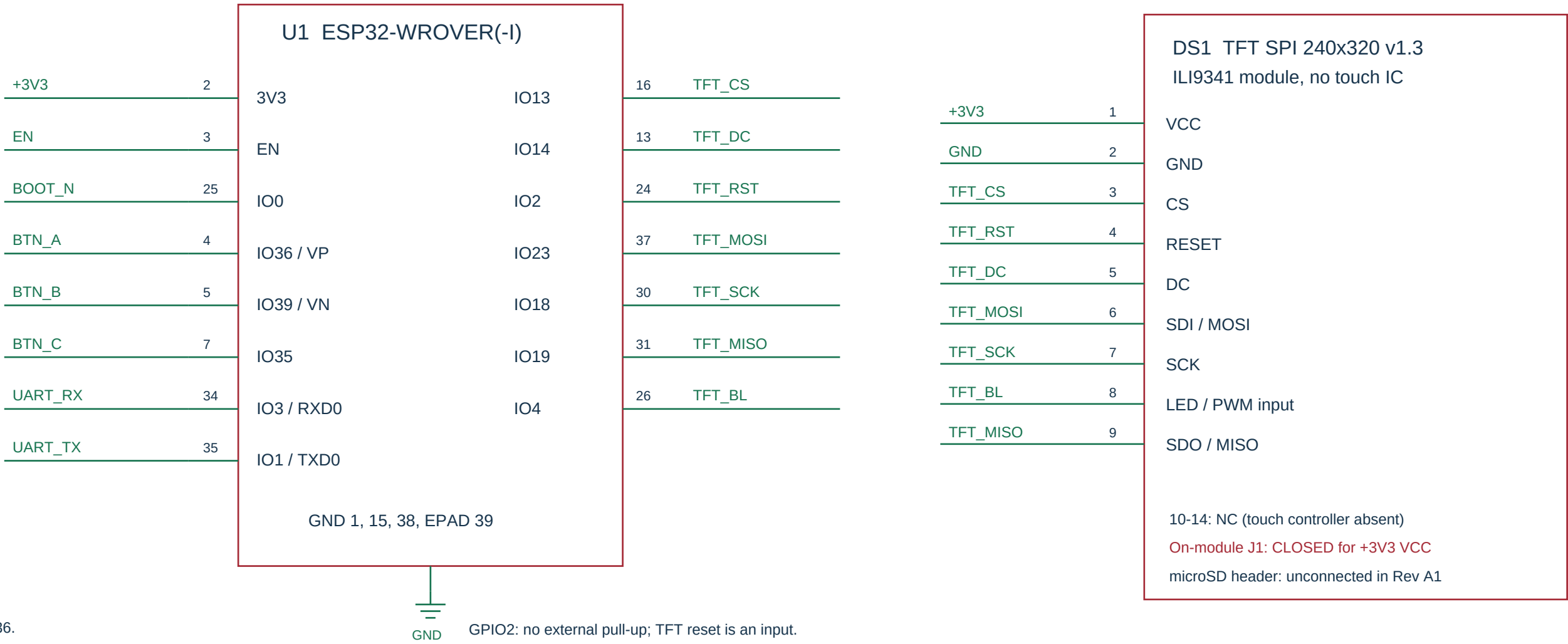
Power, controller and display

Electrical schematic | Rev A1 | 2026-09-14 | ESP32-WROVER + ILI9341 SPI 240x320

1. POWER - proposed lower-dropout supply; use a regulated 5 V / 1 A minimum source



2. SIGNALS - identical net names are electrically connected; numbers at U1 are module pad numbers



UNUSED / RESERVED U1 pads:
6, 8-12, 14, 17-23, 27-29, 32, 33, 36.
Leave unconnected in this design.

Do not use flash / PSRAM pads as GPIO.

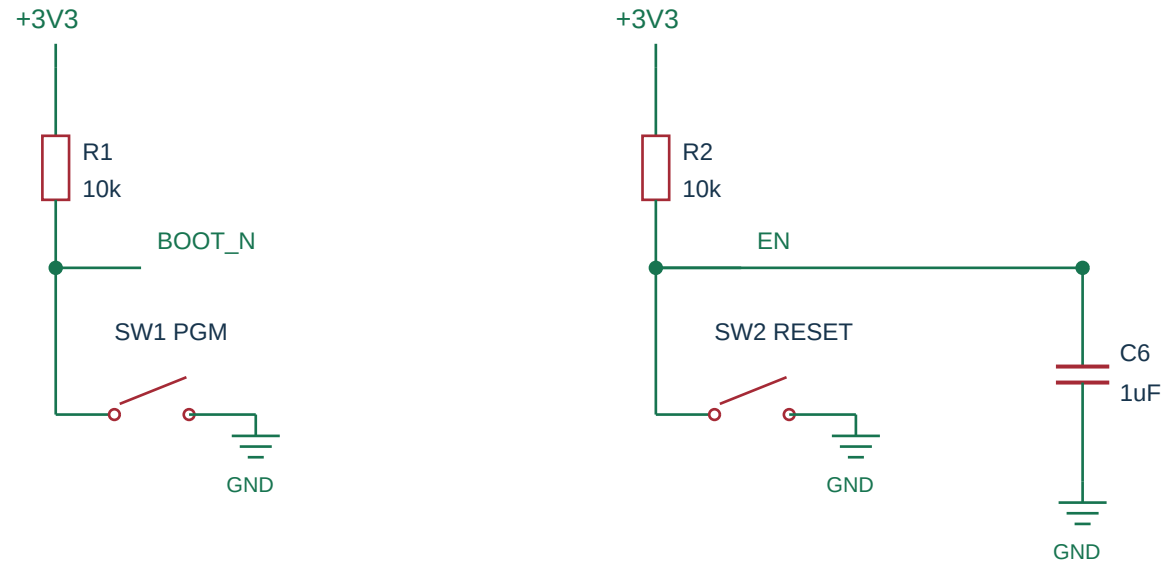
GPIO2: no external pull-up; TFT reset is an input.
GPIO12: leave unused; do not force a new boot strap.
WROVER(-I): attach the appropriate antenna before RF use.

4VRS DISPLAY

Reset, service connector and backlight

Electrical schematic | Rev A1 | 2026-09-14 | ESP32-WROVER + ILI9341 SPI 240x320

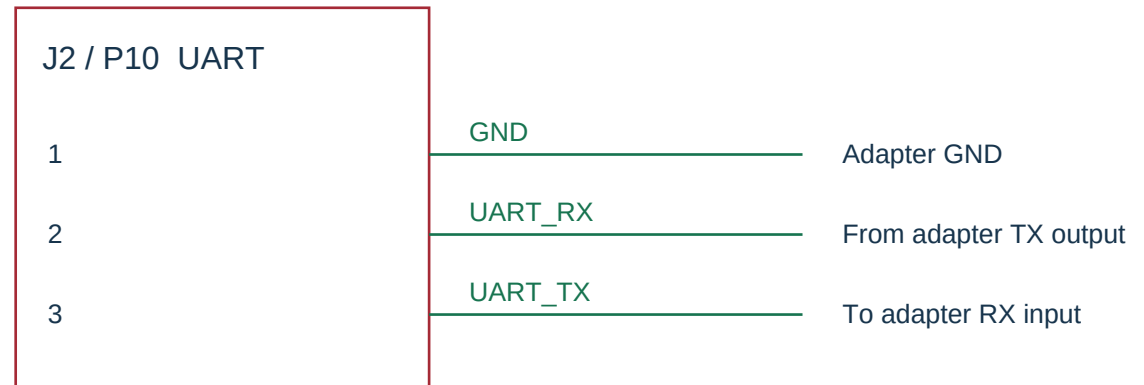
3. RESET / BOOT - added passive parts for defined startup



Hold PGM, pulse RESET, release PGM to enter the UART bootloader.

Normal start: RESET only. Do not short the EN pull-up resistor.

4. UART SERVICE - 3.3 V logic, manual reset

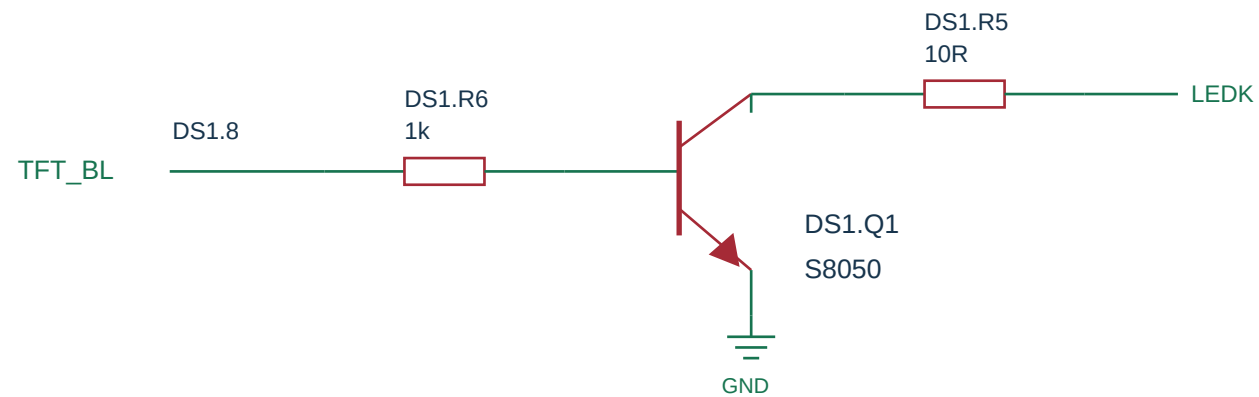


Power the device through J1; do not join two supply outputs.

Signal direction is authoritative: adapter connector labels may vary.

UART log: 115200 baud. GPIO1 = ESP TX; GPIO3 = ESP RX.

5. OUR MODULE - GPIO4 drives an existing transistor input



Equivalent circuit inside the LCDWIKI module (not extra parts).

LEDA is supplied by the module +3V3 rail.

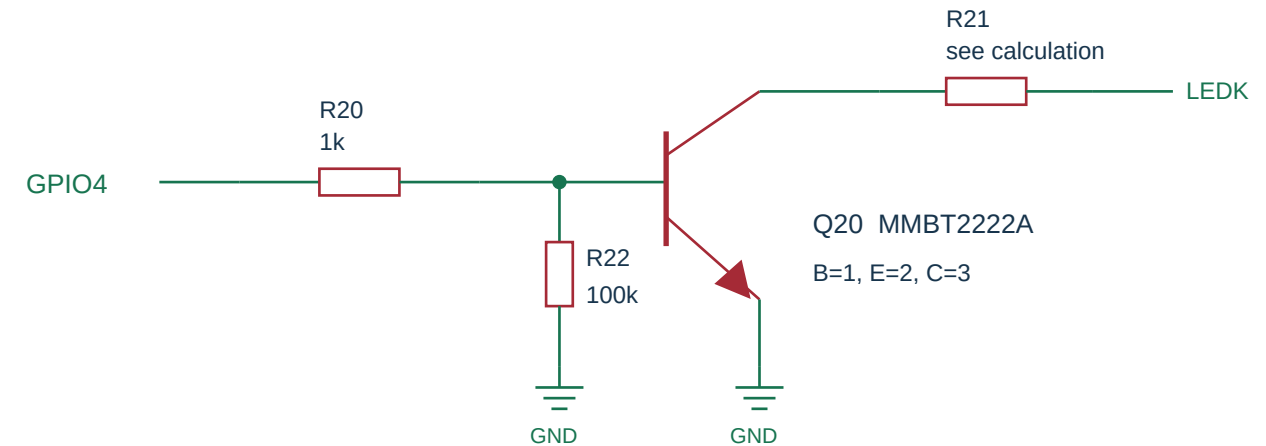
GPIO4 -> DS1 pin 8 directly; HIGH enables the backlight.

Firmware PWM: 5 kHz / 10 bit. No external NPN is fitted here.

DS1 pin 8 is a logic input, not the raw LED anode.



6. ALTERNATIVE ONLY - display with separate LEDA / LEDK



LEDA -> rated supply; LEDK -> R21 -> Q20 collector.

Example: one 3.0 V LED string, 3.3 V supply, 10 mA target.

$R21 \geq (3.3 - 3.0 - 0.1) / 0.010 = 20 \text{ ohm}$; choose 22 ohm.

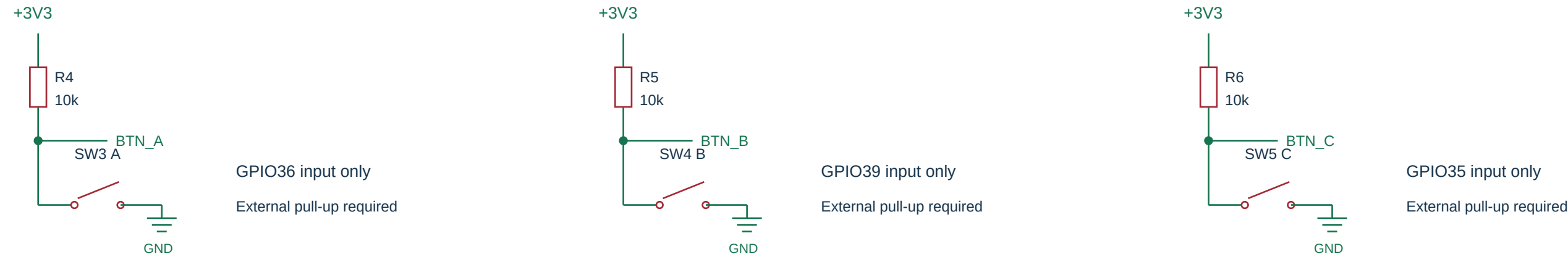
Recalculate for actual V_f , current and supply tolerances.

Parallel strings need individual limiting or a LED driver.

Not for our DS1 LED input. Do not connect both circuits.

R20 limits GPIO/base current; the original drawing omitted it.

7. OPTIONAL BUTTONS - hardware provision; navigation is not implemented in firmware 0.4.3



8. MAIN ASSEMBLY BOM

U1	ESP32-WROVER(-I), PSRAM, module
U2	AP7361C-33ER-13, 3.3 V, SOT223R
D1	1N5817, series reverse-polarity diode
J1 / J2	2-pin power / 3-pin UART connector
DS1	TFT SPI ILI9341 240x320, LED logic input
C1 / C3	10uF X7R, 10 V; effective C >= 4.7uF
C2 / C4	100nF X7R, 10 V or higher
C5	100uF electrolytic, >= 6.3 V
C6	1uF X7R, >= 6.3 V
R1 / R2	10k, 1%, 0.1 W or higher
R3	100k, 1%, optional PWM pull-down
SW1 / SW2	Normally-open momentary PGM / RESET
R4-R6 / SW3-SW5	Optional: 10k / normally-open buttons

Q20 / R20-R22 belong only to the alternative, not the main BOM.

9. BUILD CONDITIONS AND REFERENCES

U2 is a proposed change from the original L1117-33 supply.

Check the exact SOT223R package: 1=GND, 2/tab=OUT, 3=IN.

Ordinary SOT223 AP7361C has a different pinout.

Place C1/C2 close to U2 IN; C3 close to U2 OUT.

Place C4/C5 close to U1 power pins with a short ground return.

Allow at least 500 mA for ESP32 plus LCD/backlight current.

LDO heat: $P=(V_{in}-3.3)*I$; 4.7 V at 0.5 A gives about 0.7 W.

Provide copper area and check temperature inside the enclosure.

For sustained high load, use a qualified 3.3 V buck supply.

DS1 VCC=3.3 V: close its supply jumper J1 (0R bypass).

If changing DS1 VCC to 5 V, open its J1 first.

Touch pins and microSD connector are intentionally not wired.

GPIO numbers are not ESP32 chip QFN pad numbers.

No ERC/SPICE or new PCB hardware validation has been run.

Confirm supply, startup and current on the assembled hardware.

Manufacturer documentation (clickable in PDF):

LCDWIKI MSP2402: module schematic and manual

Espressif: WROVER module / hardware design guidelines

Diodes: AP7361C datasheet; original 1117: see TI datasheet